

Week 6 – Functions (Basics, No Recursion)

■ Learning Objectives

- Define and call functions to structure programs.
- Use parameters and return values to pass information.
- Understand local variables and scope.

■ Key Concepts

1. **Defining:** `def name(params):`
2. **Returning:** return value hands data back to caller.
3. **Parameters:** positional, default values, simple type hints (optional).
4. **Scope:** variables inside a function are local.

Example 1 – Simple Function

```
def greet(name):  
    return f"Hello, {name}!"  
  
print(greet("Alice"))
```

Example 2 – Compute Average

```
def average(nums):  
    if not nums:  
        return 0.0  
    total = 0  
    for x in nums:  
        total += x  
    return total / len(nums)  
  
print(average([5, 8, 10]))
```

■ Practice Problems

Practice 1 – Max of Three

Write function max3(a,b,c) returning the maximum of three numbers.

Solution:

```
def max3(a, b, c):  
    m = a  
    if b > m:  
        m = b  
    if c > m:  
        m = c  
    return m  
  
print(max3(3, 9, 5))
```

Practice 2 – Count Letters

Write count_letter(s, ch) that returns how many times ch appears in s.

Solution:

```
def count_letter(s, ch):  
    cnt = 0  
    for c in s:  
        if c == ch:  
            cnt += 1  
    return cnt  
  
print(count_letter("banana", "a"))
```

Practice 3 – Validate Password

Write is_valid(pw) that returns True if length ≥ 8 and includes a digit and a letter.

Solution:

```
def is_valid(pw):  
    has_digit = False  
    has_alpha = False  
    if len(pw) < 8:  
        return False  
    for ch in pw:  
        if ch.isdigit():  
            has_digit = True  
        if ch.isalpha():  
            has_alpha = True  
    return has_digit and has_alpha  
  
print(is_valid("abc12345"))
```

■ Homework Challenge

Write `grade(score)` returning 'A' (≥ 80), 'B' (≥ 70), 'C' (≥ 60), 'D' (≥ 50), else 'F'.

Expected Solution:

```
def grade(score):  
    if score >= 80: return 'A'  
    if score >= 70: return 'B'  
    if score >= 60: return 'C'  
    if score >= 50: return 'D'  
    return 'F'  
  
print(grade(76))
```